

MANUEL FRANA

Senior QA Automation Engineer | SDET

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SUMMARY

Senior QA Automation Engineer with 5+ years on the core backend framework an enterprise SaaS platform is built on. I identify the quality gap or architectural risk, author the technical specification, and ship the fix in production code myself — a hybrid SDET / software engineer. The platform runs on a proprietary in-house framework, so I build the test harnesses, instrumentation and release tooling rather than adopt them. Strong in test automation, API and integration testing, CI/CD quality gates, progressive rollouts and root cause analysis across PHP, Python and JavaScript backed by MySQL and Redis.

EXPERIENCE

Senior Quality Assurance Engineer — Framework Team

Oct 2022 – Present

Avature

Remote, Spain

- Own quality for the **core backend framework** — database access layer, **Redis** caching, and **metrics instrumentation** — consumed by **40** product engineering teams, combining test strategy ownership with hands-on feature development in **PHP**, **Python** and **JavaScript**.
- Independently identify coverage gaps and architectural risks, author the technical specification and acceptance criteria, then **implement and ship the change end to end** in production code — removing the handoff between QA analysis and development.
- Design and maintain **automated test suites** across **unit, integration, API and end-to-end (E2E)** layers with **Selenium WebDriver** and **custom in-house harnesses**, run by **Jenkins** pipelines that build, test and deploy the team's **Debian packages**.
- Operate **Redis** inside **Docker** containers as part of the framework stack, handling container configuration and runtime operations such as enabling **authentication** on running instances.
- Re-architected monolithic code-style checks into **module-scoped, independently schedulable units**, unlocking **parallel execution** and cutting suite runtime by up to **50%**.
- Own the **progressive rollout plan** for a new framework feature — enabling it incrementally across client instances and **measuring performance and production impact** at each stage before wider release.
- Automate server-side operational procedures with **Rundeck** jobs for **self-service execution**, and coordinate a **weekly working session with 2 QA engineers** to prioritize testing and automation work in an **Agile/Scrum** environment.

Quality Assurance Engineer

Jun 2021 – Oct 2022

Avature

Remote, Argentina

- Built **automated regression and E2E test suites** with **Selenium** and unit tests, replacing **20** recurring manual regression passes per release cycle.
- Executed **exploratory, data consistency and regression testing** across releases, authoring reproducible **defect reports** through to resolution, and validated backend behaviour by inspecting **HTTP traffic** and querying **MySQL** directly to confirm **data integrity**.

PROJECTS

- Catox** — Multiplayer Strategy Game | *Luau, React (jsdotlua), Rodux, Rojo, Wally, TestEZ* 2026 – Present
- Architected a **server-authoritative multiplayer game** (~49K LOC Luau, live on Roblox) where every state transition is validated server-side, routing all RemoteEvents and RemoteFunctions through a single **Command pattern gateway** so validation, authorization and abuse detection live in one layer rather than hundreds of call sites.
 - Wrote core game logic as **headless modules with injected dependencies** so it could be mocked and run outside the Roblox engine — a **67-spec automated suite** (TestEZ + Lune) over logic the engine cannot unit test. Abstracting execution from any real player session also made **AI opponents** possible.
- Courts** — Booking Automation Bot | *Python, Playwright, asyncio, SQLite, Docker* 2026
- Built an **asynchronous Telegram bot** that scrapes live tennis court availability and **auto-reserves** slots the moment they are released at midnight, automating a booking race a group of players kept losing.
 - Reverse-engineered an ASP.NET booking platform's internal APIs (ViewState / EventValidation, session-bound tokens) and drove authenticated flows with **Playwright** and **httpx**; encrypted credentials at rest with **Fernet (AES)** and persisted sessions, reservations and cached availability in **SQLite**.
 - Diagnosed a production outage through **root cause analysis** — booking tokens fetched in one HTTP session were rejected when submitted from another — shipped the same-session fix and documented it in a written **post-mortem**. **Dockerised** deploy with a persistent volume; **pytest** coverage over the availability parser.

TECHNICAL SKILLS

Languages: PHP, Python, JavaScript/TypeScript, Luau, Java, SQL, HTML/CSS, Bash

Testing & QA: Test Automation Frameworks, Selenium WebDriver, Playwright, pytest, Unit / Integration / API / End-to-End (E2E) Testing, Regression Testing, Exploratory Testing, Parallel Test Execution, Test Plans & Cases, Defect Tracking, Root Cause Analysis, Shift-Left Testing, Postman, Charles Proxy

Backend & Data: REST APIs, MySQL, Redis, Caching Strategies, SQLite, Database Design, Data Consistency Validation

DevOps: Docker, Jenkins, CI/CD Pipelines, Debian Packaging, Rundeck, Git & GitHub, Linux, Metrics Instrumentation, Log Analysis, Monitoring & Alerting

Practices: Agile / Scrum, SDLC & STLC, Release Management, Progressive / Phased Rollouts, Code Review, Technical Specification Writing, Cross-Functional Collaboration, Mentoring

EDUCATION

Universidad Tecnológica Nacional (UTN) — Córdoba Graduated Jun 2024
Engineer's Degree, Information Systems Engineering (5-year program) — GPA 8.67 / 10 Córdoba, Argentina

Cambridge University — First Certificate in English (FCE) England

LANGUAGES

Spanish — Native | English — Advanced (professional working proficiency)